MNIST-CNN CLASSIFICATION

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers

print("TensorFlow version:", tf.__version__)



(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()

print("Training images:", x_train.shape)
print("Training labels:", y_train.shape)
print("Test images:", x_test.shape)
print("Test labels:", y_test.shape)

plt.figure(figsize=(10, 5))

for i in range(10):
    plt.subplot(2, 5, i + 1)
    plt.imshow(x_train[i], cmap="gray")
    plt.title(f"Label: {y_train[i]}")
    plt.axis("off")

plt.tight_layout()
plt.show()

x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0

x_train = np.expand_dims(x_train, axis=-1)
x_test = np.expand_dims(x_test, axis=-1)

print("Training shape:", x_train.shape)
print("Testing shape:", x_test.shape)

model = keras.Sequential([


    layers.Input(shape=(28, 28, 1)),


    layers.Conv2D(32, (3, 3), activation="relu"),


    layers.MaxPooling2D((2, 2)),


    layers.Conv2D(64, (3, 3), activation="relu"),
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    layers.MaxPooling2D((2, 2)),

    layers.Flatten(),

    layers.Dense(128, activation="relu"),

    layers.Dropout(0.5),

    layers.Dense(10, activation="softmax")
])

model.summary()

from tensorflow.keras import layers, models

model = models.Sequential([

    layers.Conv2D(32, (3, 3), activation="relu", input_shape=(32, 32, 3)),

    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(64, (3, 3), activation="relu"),
    layers.MaxPooling2D((2, 2)),

    layers.Flatten(),

    layers.Dense(128, activation="relu"),

    layers.Dropout(0.5),

    layers.Dense(10, activation="softmax")
])

model.compile(optimizer="adam",
              loss="categorical_crossentropy",
              metrics=["accuracy"])

model.summary()
```